

Activity 2

1) Stack operation:

push(54) $\rightarrow$ stack: [54]

push(52) $\rightarrow$ [54, 52]

pop() $\rightarrow$ pops 52, [54]

push(55) $\rightarrow$ [54, 55]

push(62) $\rightarrow$ [54, 55, 62]

s = pop() $\rightarrow$ s = 62

Queue operation:

enqueue(21) $\rightarrow$ [21]

enqueue(24) $\rightarrow$ [21, 24]

dequeue() $\rightarrow$ [24]

enqueue(28) $\rightarrow$ [24, 28]

enqueue(32) $\rightarrow$ [24, 28, 32]

q = dequeue() $\rightarrow$ q = 24

s + q = 62 + 24 = 86

2)

10 5 + 60 6 / 8

3)

1) 10 5 + $\rightarrow$ 10 + 5 = 15

stack = [15]

2) 60 6 / $\rightarrow$ 60 / 6 = 10

stack = [15, 10]

3) 15 10 * $\rightarrow$ 15 * 10 = 150

4) 150 2 - $\rightarrow$ 150 - 2 = 148

4)

Minimum no of enqueue is 0

Elements can be moved from Q₁ to Q₂ by repeatedly dequeuing from Q₁ and enqueueing to Q₂

5)

Modified stack with REVERSE can be implemented using a queue and can implement a FIFO queue with O(1) time per operation

$$b) \quad 8 \quad 2 \quad 3^{\wedge} / 2 \quad 3^* + 5^{\wedge}$$

Ans

$$2 \quad 3^{\wedge} = 8$$

$$8 \quad 8 / = 1 \quad : \quad \text{stack} = [1]$$

$$\text{push } 2, 3 \rightarrow \text{stack} = [1, 2, 3]$$

$$2 \quad 3^* = 6 \rightarrow \text{stack} = \boxed{[6, 1]}$$

c)

$$a + b * c - d^{\wedge} e^{\wedge} f$$

$$b * c = bc^*$$

$$a + bc^* = abc^* +$$

$$e^{\wedge} f = ef^{\wedge}$$

$$d^{\wedge} ef^{\wedge} = def^{\wedge\wedge}$$

$$abc^* + - def^{\wedge\wedge} = abc^* + def^{\wedge\wedge} -$$

d)

$$\text{push}(10) \rightarrow \text{stack} = [10]$$

$$\text{push}(20) \rightarrow [10, 20]$$

$$\text{pop}() \rightarrow [10]$$

$$\text{push}(10) \rightarrow [10, 10]$$

$$\text{push}(20) \rightarrow [10, 10, 20]$$

$$\text{pop} \rightarrow [10, 10]$$

$$\text{pop} \rightarrow [10]$$

$$\text{pop} \rightarrow []$$

$$\text{push}(20) \rightarrow [20]$$

$$\text{pop} \rightarrow []$$